
Report: **AM2024-0074**

SEM & BET investigation on particle samples of natural origin

Short Summary 3

Client:

Curia Germany GmbH
Industriepark Höchst D569
65926 Frankfurt am Main

Contractor:

Fraunhofer-Institut für Silicatforschung ISC
Zentrum für Angewandte Analytik ZAA
Neunerplatz 2
97082 Würzburg
Germany

Project Management:

Dr. Alexander Reinholdt
Head of Center for Applied Analytics

Documentation Overview

Contact

on the part of the client

Hannah Sauer and Dr. Torsten Busch

on the part of the contractor

Dr. Alexander Reinholdt

Phone: +49 931 4100-260

E-Mail: alexander.reinholdt@isc.fraunhofer.de

ISC Offer Number

2210-2024-51000-45-02801-00010 and
2210-2024-51000-45-02801-00019

Order of the Client

signature on offer 2210-2024-51000-45-02801-00010
dated 26 July 2024 and
signature on offer 2210-2024-51000-45-02801-00019
dated 09 September 2024

Place of Performance and Processing Time

Place of Performance

Center for Applied Analytics
Fraunhofer Institute for Silicate Research
Neunerplatz 2
97082 Würzburg
Germany

Processing Time

Sample receipt	28.08.2024
Begin of the analysis	19.09.2024
End of the analysis	ongoing
Date of report	23.09.2024

Sample Overview

Sample No.	Description
P03	Graphene – Batch: MB2976A – Date: 23.08.2024

P01

Procedure of Testing

- The powder sample was used as received for the BET N₂ sorption measurements as well as the SEM measurements.
- To determine the surface area, the powder was measured in a Quantachrome Autosorb device. The amount of the analysed material was quite little ($m = 0.0194$ g), so the results should be interpreted with a grain of salt.
- For SEM imaging, the powder was applied to a carbon pad and then imaged without further treatment (no sputter coating applied).

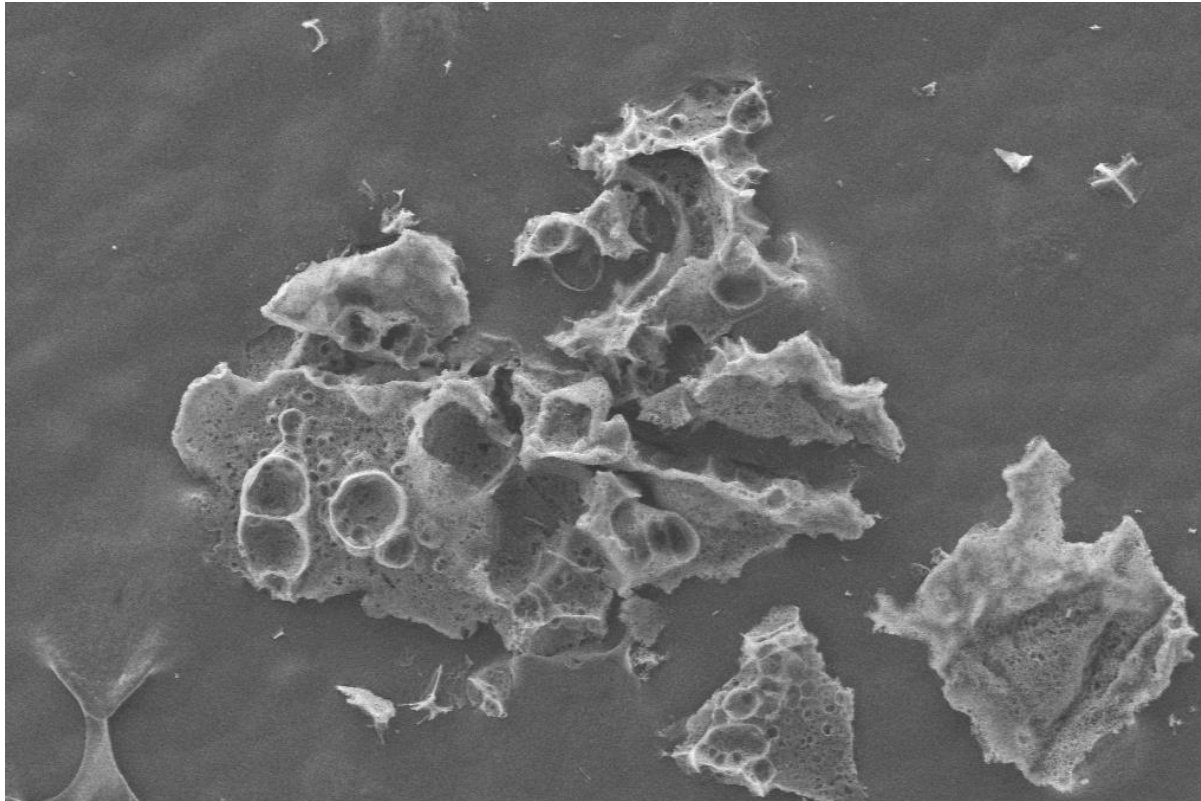
BET Surface Area

Sample P03 (MB2976A)

Sample No.	BET Surface Area [m ² /g]	Langmuir Surface Area [m ² /g]
P03	1.88 x 10 ³	2.55 x 10 ³

SEM investigation

Sample P03 (MB2976A) – Area 1



100 μ m
Mag = 100 X

Signal A = SE2
EHT = 15.00 kV
WD = 8.8 mm

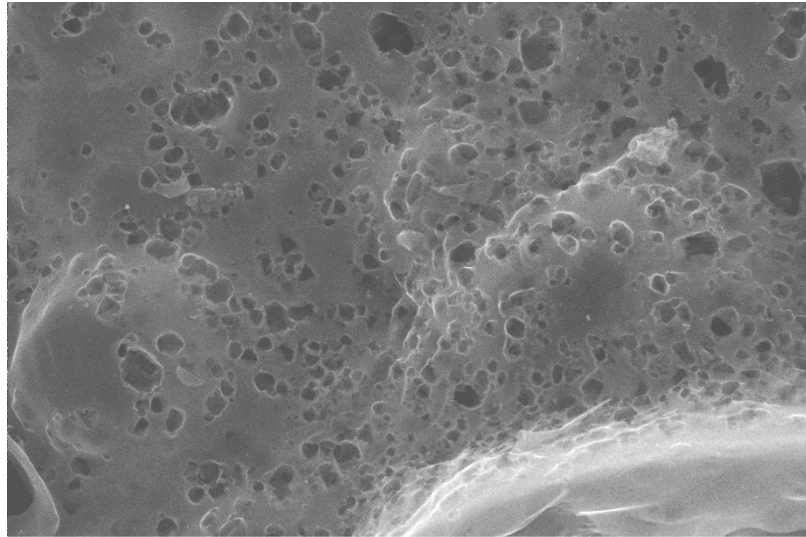
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_02.tif

AM2024-0074 P03
Date :22 Sep 2024
stck; ZA

- The image shows several big, porous particles on the carbon pad at a low magnification (100 times).
- Their sizes are in the range of 100 μ m and bigger.

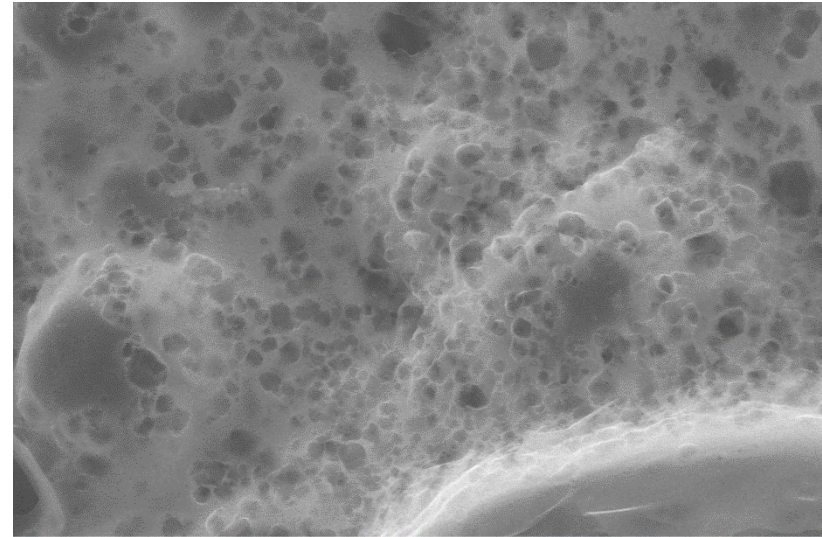
SEM investigation

Sample P03 (MB2976A) – Area 1



2 μ m
Mag = 2.00 K X
Signal A = InLens
EHT = 15.00 kV
WD = 8.8 mm
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_03.tif

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Date :22 Sep 2024
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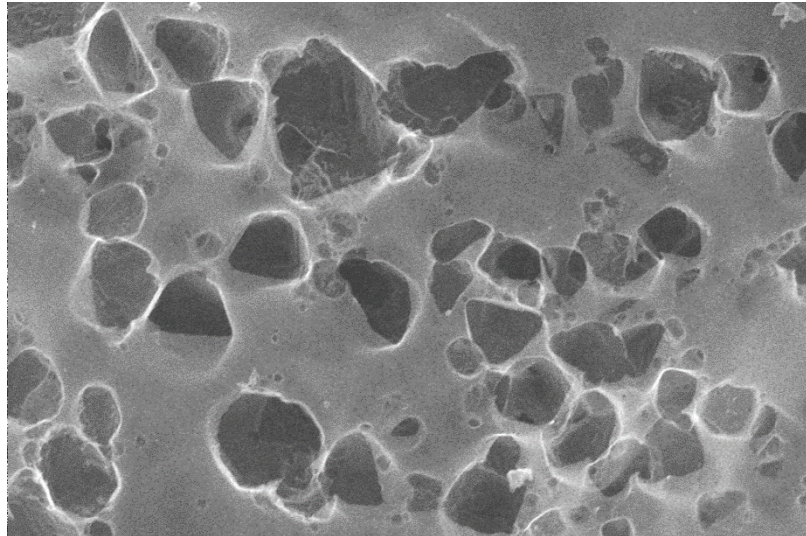
2 μ m
Mag = 2.00 K X
Signal A = SE2
EHT = 15.00 kV
WD = 8.8 mm
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_04.tif

AM2024-0074 P03
Date :22 Sep 2024
stck; ZA

- These images show one and the same area at a magnification of 2,000 times taken with two different detectors.
- The *inlens* detected emphasises edges (left), and the sample is imaged with high surface sensitivity.
- The *SE2* detector (right) emphasises the topography, and the image contains information from greater depths of the sample, which creates a translucent effect.

SEM investigation

Sample P03 (MB2976A) – Area 1

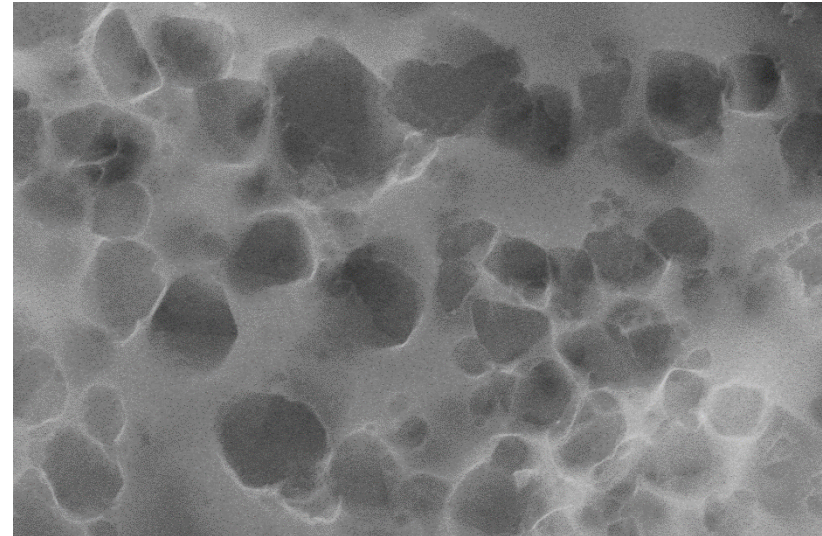


1 μm
Mag = 10.00 K X

Signal A = InLens
EHT = 15.00 kV
WD = 8.8 mm

Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_07.tif

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1 μm
Mag = 10.00 K X

Signal A = SE2
EHT = 15.00 kV
WD = 8.8 mm

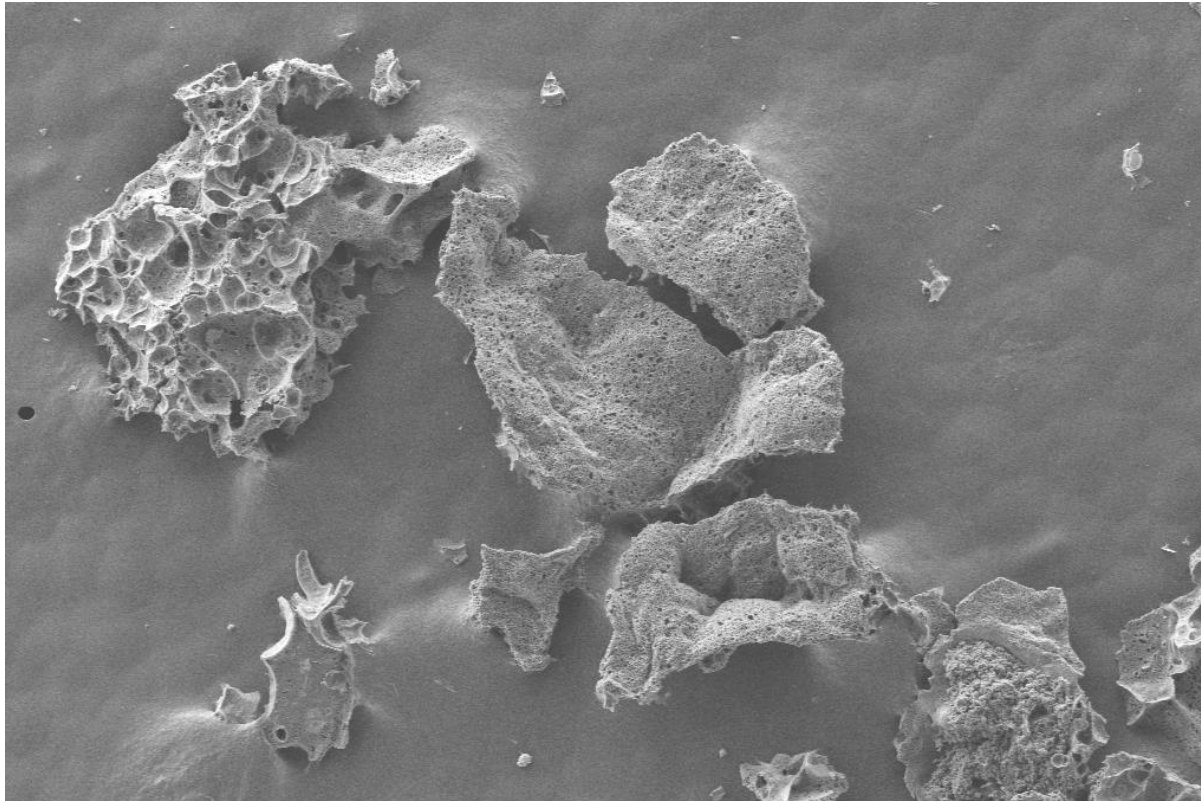
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_08.tif

AM2024-0074 P03
Date :22 Sep 2024
stck; ZA

- These images show a part of the aforementioned area at a higher magnification of 10,000 times.
- The images were again taken with the two different detectors (left *inlens* and right *SE2*).

SEM investigation

Sample P03 (MB2976A) – Area 2



100 μm
Mag = 100 X

Signal A = SE2
EHT = 5.00 kV
WD = 4.9 mm

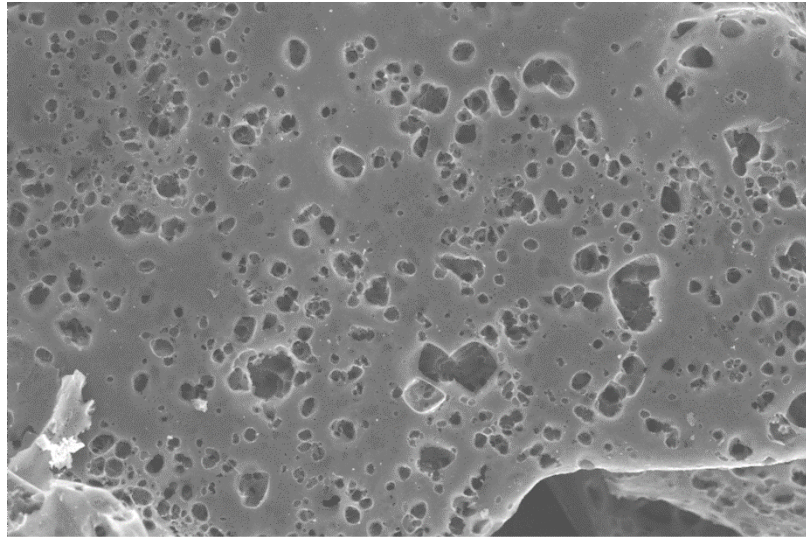
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_12.tif

AM2024-0074 P03
Date :22 Sep 2024
stck; ZA

- The image shows several big, porous particles in a different area of the carbon pad at a low magnification (100 times).
- Their sizes are in the range of 100 μm and bigger.

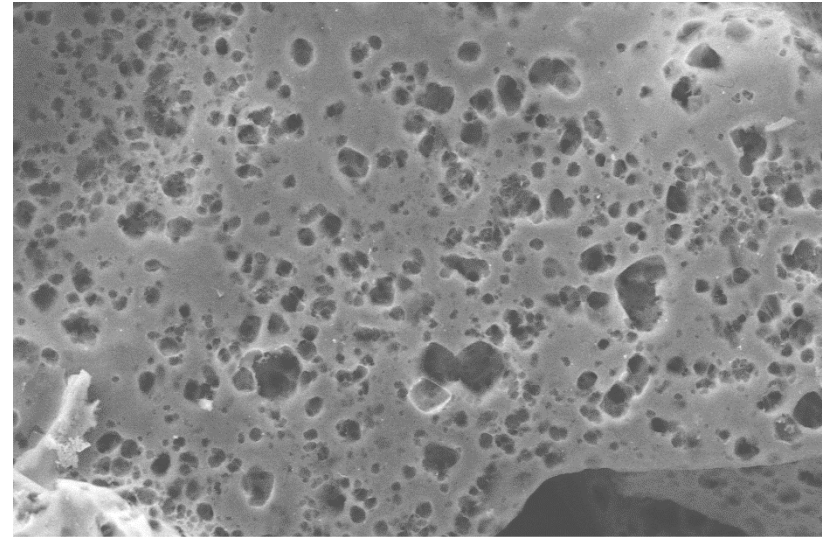
SEM investigation

Sample P03 (MB2976A) – Area 2



2 μ m
Mag = 2.00 K X
Signal A = InLens
EHT = 5.00 kV
WD = 4.9 mm
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_19.tif

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stck; ZA



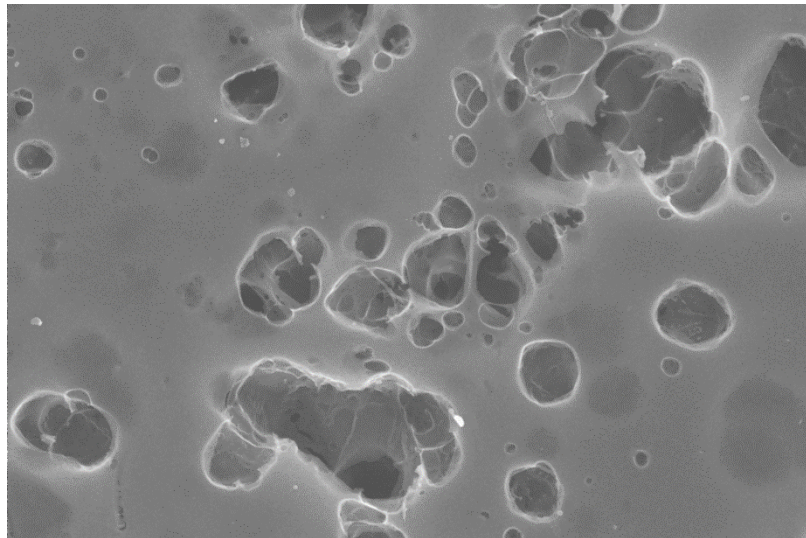
2 μ m
Mag = 2.00 K X
Signal A = SE2
EHT = 5.00 kV
WD = 4.9 mm
Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_20.tif

AM2024-0074 P03
Date :22 Sep 2024
stck; ZA

- These images show one and the same area at a magnification of 2,000 times taken with two different detectors.
- The *inlens* detected emphasises edges (left), and the sample is imaged with high surface sensitivity.
- The *SE2* detector (right) emphasises the topography, and the image contains information from greater depths of the sample, which creates a translucent effect in some parts of the image. Page 11

SEM investigation

Sample P03 (MB2976A) – Area 2

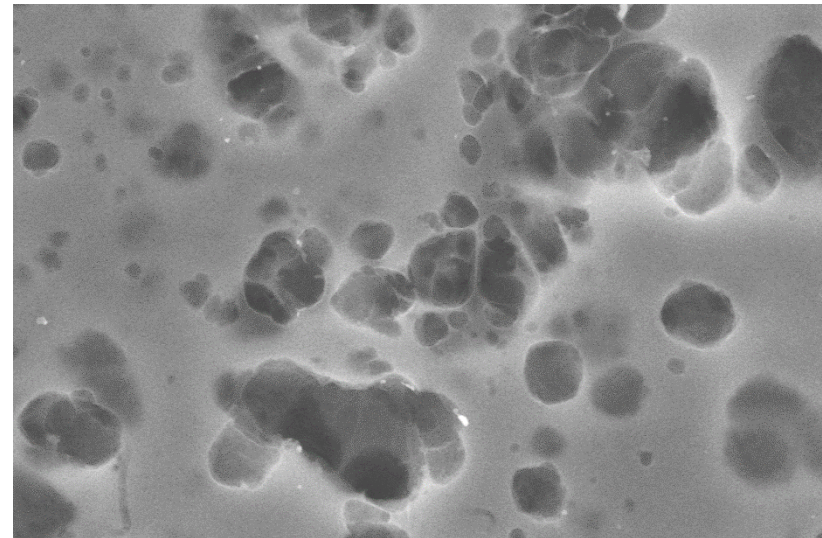


1 μm
Mag = 10.00 K X

Signal A = InLens
EHT = 5.00 kV
WD = 4.9 mm

Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_21.tif

AM2024-0074 P03
Date :22 Sep 2024
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1 μm
Mag = 10.00 K X

Signal A = SE2
EHT = 5.00 kV
WD = 4.9 mm

Batch: MB2976A (original, not crushed)
Top View
File Name = AM2024-0074_P03_TV-2_22.tif

AM2024-0074 P03
Date :22 Sep 2024
stck; ZA

- These images show a part of the aforementioned area at a higher magnification of 10,000 times.
- The images were again taken with the two different detectors (left *inlens* and right *SE2*).

Final Remarks

- The analyses are carried out by specifically qualified employees and by the use of up-to-date analysis machines. Nevertheless, we are not able to provide guarantee for the correctness of the drawn conclusion.
- The results only refer to the provided and examined samples.
- Sample taking and shipping of the samples to the ZAA took place under the responsibility of the client.
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